

to do with adversity—the word stress was notably used in engineering to describe the effect on materials.

Since then, there have been many studies that measure both the positive and negative effect of stress on our bodies and emotional states, such as the study conducted in 2015 by Silvia Maier, Aidan Makwana, and Todd Hare reported as ‘Acute Stress Impairs Self-Control in Goal-Directed Choice by Altering Multiple Functional Connections within the Brain’s Decision Circuits’. The name of the study kind of gives the ending away, but it’s of interest to us here because it looks at stress, temptation and our old friend, self-control.

In the study, the researchers tested 22 participants who were chosen because they wanted to maintain a healthy lifestyle but struggled with eating junk food and therefore faced day to day self-control challenges. The researchers split the participants into two groups—the stress and control groups. The difference between the two groups was that during the three hours the participants spent in the lab, the stress group had to immerse their hand in an ice water bath for three minutes while being videotaped. They were told not to communicate with the videographer and while they could remove their hand from the ice bath if it became too cold, they were not allowed to look away from the camera. The control group meanwhile had the same experience except they had to keep their hand in a warm water bath for the same length of time and the experiment wasn’t taped.

I don’t know about you, but I can guess the result. The majority of the stress group reached for the unhealthy food option to reward themselves for going through their ordeal. The researchers found that

stressed participants’ choices were more affected by short-term taste reward and that they encoded taste more strongly in portions of the amygdalae and ventral striatum. Furthermore, the stress manipulation increased task-dependent connectivity between these limbic regions and a portion of the ventromedial prefrontal cortex (vmPFC) that represented integrated stimulus value ... In addition, increased stress levels were associated with decreased connectivity between vmPFC and dorsolateral prefrontal cortex (dlPFC) regions that were activated when engaging self-control.